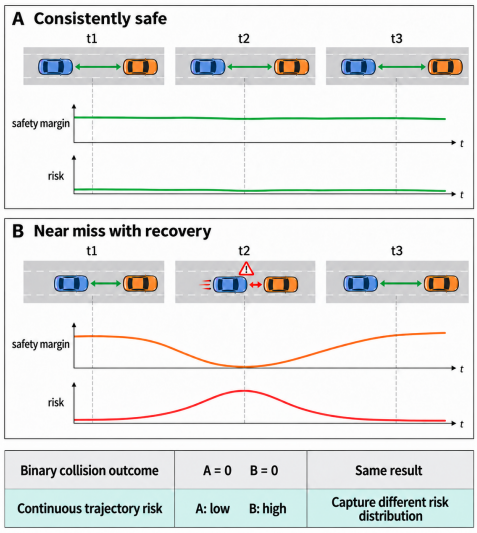
Which Trajectory Is Safer? Learning Future Risk for Autonomous Driving

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***Abstract*— Collision rate remains the dominant metric for evaluating trajectory safety, yet as planners increasingly avoid collisions outright, it reveals little about the safety differences among trajectories that remain collision-free.**

We introduce FutureRisk, a metric that scores the *latent future risk* of a trajectory from its own realized future interactions, whether logged from human driving or rolled out by a policy in closed-loop simulation, producing a continuous score that distinguishes collision-free trajectories by how closely and how persistently they approached an unsafe interaction state. At its core, a Context-Adaptive Risk Model (CARM) replaces fixed, hand-crafted TTC-style thresholds with a learned, context-dependent safety criterion, while remaining bounded around an interpretable, closed-form geometric prior. The resulting score is continuous, interpretable, and directly comparable across trajectories on matched scenarios, enabling fine-grained safety evaluation beyond binary collision outcomes.



I. Introduction

Recent planning systems have achieved substantial improvements in collision avoidance, resulting in increasingly low collision rates [1]. Nevertheless, collision occurrence, which often summarized as a collision rate, remains a dominant safety measure. This binary outcome reveals little about the safety differences among trajectories that ultimately remain collision-free [7] and may therefore produce overly optimistic assessments by obscuring risks that do not materialize as collisions [2], [3]. In practice, a collision-free trajectory is not necessarily a sufficiently safe one: some trajectories maintain ample safety margins throughout an interaction, whereas others enter safety-critical situations with limited room for recovery or avoid collisions only because surrounding agents respond favorably. Collision-based evaluation cannot distinguish among these substantially different safety profiles.

To move beyond binary collision outcomes, prior work has introduced prospective safety and criticality criteria that quantify proximity to hazardous interactions [9]. However, existing criteria exhibit a persistent trade-off between two desirable properties: interpretability and context adaptivity. Formula-based criteria are interpretable by construction, but typically rely on fixed assumptions and thresholds that may not adequately reflect differences across driving contexts [4]. Learning-based alternatives can adapt to contextual variations, but often lack the explicit closed-form structure that makes their outputs physically interpretable [10]. To our knowledge, existing criteria do not explicitly combine both properties within a unified metric.

Together, these gaps motivate us to reconsider how the safety of an executed trajectory should be characterized beyond its eventual collision outcome. A more informative safety metric should not only distinguish different levels of risk among collision-free trajectories, but also remain interpretable while adapting its safety criterion to the interaction context. Drawing on prior notions of prospective safety and criticality [8], we characterize risk at each point along an executed trajectory by how closely and how persistently its subsequent evolution approaches an unsafe interaction state. This risk is *future* with respect to the evaluated point, but is computed retrospectively from the future evolution that actually occurred. It is *latent* because it may remain hidden when the interaction does not ultimately result in a collision. We therefore refer to this quantity as **latent future risk**. Quantifying latent future risk places collision-free trajectories along a continuous spectrum of safety, addressing the limitations of binary collision outcomes; doing so through an interpretable yet context-adaptive criterion further resolves the trade-off exhibited by existing criticality measures. This leads to the central question of this work: **How can the latent future risk of an executed trajectory be quantified in a manner that is both interpretable and adaptive to scenario context?**

To this end, we propose **FutureRisk**, a metric that quantifies the latent future risk of an executed trajectory from its realized future interactions. At its core, our **Context-Adaptive Risk Model (CARM)** starts from a geometry-aware 3D time-to-collision (TTC), which provides a closed-form physical measure of geometric criticality, and learns how the corresponding safety criterion should vary with the observed interaction context. CARM thereby replaces universal TTC thresholds and hand-crafted scenario rules with a context-dependent criterion, while retaining the closed-form measure as a physical anchor for interpretation. FutureRisk uses this criterion to capture both the severity and persistence of risk along the realized trajectory. The resulting continuous score

reveals fine-grained safety differences among trajectories that share the same collision outcome, supporting trajectory-level safety auditing as well as method-level comparison through score distributions over matched scenarios.

Our main contributions are summarized as follows:

- We introduce **FutureRisk**, a continuous trajectory-level safety metric that assesses risk from interactions with surrounding agents. Beyond binary collision outcomes, FutureRisk provides a fine-grained assessment of interaction risk to distinguish trajectories that remain collision-free but exhibit different levels of safety.
- We develop the **CARM**, which combines critical-event timing with a context-dependent safety-time distribution whose median is constrained around a closed-form physical prior. A two-stage training scheme first learns this structured risk estimate and then fits a bounded logit residual with the structured branch frozen, allowing further adjustment while explicitly limiting its effect on the final score.
- We evaluate FutureRisk through its agreement with collision outcomes, its discrimination among collision-free trajectories with different safety margins, and its zero-shot transfer across datasets. Qualitative analyses further examine how the metric responds to observed interaction dynamics.

II. Related Work

Beyond collision rate. Recent planning systems have driven collision rates increasingly low [1], which has sharpened rather than resolved a long-standing concern: collision occurrence is a direct measure of safety failure, but as an outcome-based metric it reveals little about the safety differences among trajectories that ultimately remain collision-free [2], [3]. Prior work on real-time safety metrics has argued directly that observing eventual collisions or near-collisions is insufficient to evaluate whether an automated driving system maintains a continuously safe operating state [7]. This coarseness persists even in more sophisticated evaluation frameworks: nuPlan's Closed-Loop Score blends progress, comfort, and minimum TTC into a continuous value, yet still gates that value multiplicatively to zero on any at-fault collision or off-road event, so its underlying safety judgment collapses back to a binary pass/fail the moment a violation occurs [6]. The Waymo Open Sim Agents Challenge shows this can be exploited outright: a trivially static policy can drive collision rate to zero without being safe [5]. Together, these findings motivate safety metrics that assign continuous, fine-grained scores to trajectories rather than relying on sparse collision outcomes.

Rule-based safety criteria. One response is to define safety through closed-form, rule-based criteria over kinematic or geometric quantities, such as time-to-collision (TTC), post-encroachment time, and deceleration rate to avoid a crash [9], [8], or explicit safety envelopes such as Responsibility-Sensitive Safety [11]. Because these criteria are given by an explicit formula, they are transparent, auditable, and reliable in the sense that their behavior can be verified independently of any training data. However, their thresholds and assumptions are hand-specified once and then applied uniformly, without regard for the specific scene or object being evaluated. Westhofen *et al.*'s systematic review of criticality metrics shows the resulting cost: no single formulation is valid across all scenario types, and even a continuous measure such as TTC is effective for car-following but breaks down at intersections, where single-point kinematics no longer characterize the interaction [4]. Rule-based criteria therefore trade flexibility for reliability and interpretability.

Data-driven safety criteria. A complementary line of work learns safety criteria directly from trajectory data instead of specifying them by hand. Lu *et al.* embed sequences of surrogate safety measures such as TTC into a latent space with unsupervised representation learning and identify traffic conflicts by clustering that space, allowing the resulting criterion to reflect patterns in the data rather than a fixed threshold [10]. This flexibility comes at the cost of the reliability and interpretability that rule-based criteria offer: the learned criterion no longer corresponds to a verifiable closed-form quantity, and because it is fit to the interaction patterns present in its training data, its behavior on scenes or objects that differ from that distribution is not guaranteed, making generalization to unseen or underrepresented scenarios a central concern for this line of work. FutureRisk is designed to sit between these two extremes: its Context-Adaptive Risk Model learns a scene- and object-dependent safety criterion, as in this line of work, while remaining bounded around the same closed-form geometric quantities used by rule-based criteria, so that the resulting risk score remains anchored to an interpretable formula even as it adapts to context.

III. Method

A. Overview

FutureRisk assigns a continuous risk score to an ego trajectory based on its interactions with surrounding agents. Let $\boldsymbol{\xi}_e$ denote an ego trajectory over a horizon H, and let $\mathcal{W}^{\mathrm{obs}} = \{\boldsymbol{\xi}_j\}_{j=1}^{J}$ denote the trajectories of J surrounding agents over the same horizon. FutureRisk maps these inputs to a continuous score

$$R = f_{\mathrm{FR}}\big(\boldsymbol{\xi}_e, \mathcal{W}^{\mathrm{obs}}\big) \in [0,1], \tag{1}$$

where a larger R means the trajectory approached an unsafe interaction state more closely, or stayed near one for longer. A binary collision label assigns the same value to a trajectory that kept a wide margin throughout and to one that avoided contact by a fraction of a second. But R separates them. Throughout, time within the horizon is discretized into K bins with edges $0 = b_0 < b_1 < \cdots < b_K = H$; every time-dependent quantity introduced below is evaluated on this one grid, and $\sigma(\cdot)$ always denotes the logistic function.

At the core of f_{FR} is CARM: a risk prediction model with a learned, context-dependent safety criterion, while remaining bounded around an interpretable, closed-form physics prior. Concretely, for each surrounding agent j, CARM corrects a structured risk estimate $\widetilde{R}^{j}_{\mathrm{CARM}}$ with a further,

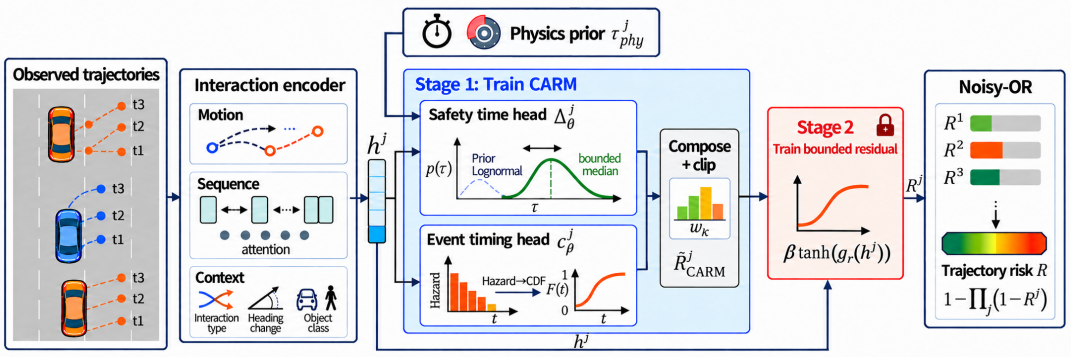


Fig. 1. Overview of how FutureRisk is computed.

learned bounded correction to obtain a per-agent risk R^j (Sec. III-C), and f_{FR} aggregates these across all agents (Sec. III-D):

$$R = 1 - \prod_j \left(1 - R^j\right),$$
$$R^j = \sigma\left(\mathrm{logit}\left(\widetilde{R}^j_{\mathrm{CARM}}\right) + \underbrace{\beta\tanh\left(g_r(\boldsymbol{h}^j)\right)}_{\text{learned, Stage 2}}\right). \tag{2}$$

Here $\mathrm{logit}(\cdot)$ denotes the inverse of $\sigma(\cdot)$; $\boldsymbol{h}^j$ is a shared representation of the interaction between the ego and agent j, detailed in Sec. III-B; g_r is a learned scalar head reading from $\boldsymbol{h}^j$; and β is a fixed constant bounding the magnitude of this Stage-2 correction. $\widetilde{R}^j_{\mathrm{CARM}}$ is itself built from the two arguments that Γ combines: the first argument is a required-safety-margin term, namely one raw, closed-form physics quantity plus one learned correction to it; the second argument is a separately learned, unanchored quantity for *when* the interaction turns critical (Sec. III-B):

$$\widetilde{R}^j_{\mathrm{CARM}} = \Gamma\Big(\underbrace{\tau^j_{\mathrm{phy}}}_{\text{raw}} + \underbrace{\Delta^j_\theta}_{\text{learned, Stage 1}}, \underbrace{c^j_\theta}_{\text{learned, Stage 1}}\Big). \tag{3}$$

τ^j_{phy} is the one raw, closed-form quantity in this composition: a physics prior for the required safety margin, summarizing closing speed, available deceleration, reaction time, and prediction uncertainty. It is distinct from the geometry-aware 3D-TTC τ^j_{3D}, which never enters CARM's *forward computation* at all; τ^j_{3D} is used only, offline, to construct the training target that supervises the whole model, as Sec. III-E makes precise. Every other quantity above is a network output. Δ^j_θ is a bounded correction to τ^j_{phy}. c^j_θ is a separate, unanchored term for *when* the interaction turns critical, realized as a discrete-time hazard distribution that predicts the probability the interaction has already turned critical by each time bin; it is supervised independently, so that criticality is never defined circularly in terms of the very margin it is combined with. Γ composes the two into a single structured estimate. The Stage-2 correction $\beta\tanh(g_r(\boldsymbol{h}^j))$ from Eq. (2) then applies a further, equally free correction on top of it, trained only after the rest of the network is frozen. Sec. III-E derives the geometric target that supervises R_θ, the score R from Eq. (2), viewed as a function of the model's trainable parameters θ, together with the independent label that supervises c^j_θ, and the two-stage schedule that trains every piece above.

B. CARM Architecture

CARM turns the interaction between the ego and agent j into the structured estimate $\widetilde{R}^j_{\mathrm{CARM}}$, which is later corrected by a further bounded residual (Sec. III-C). Building $\widetilde{R}^j_{\mathrm{CARM}}$ takes four steps: a shared interaction representation, a physically anchored model of how much safety time the interaction requires, an independently learned model of when it turns critical, and a composition of the two into a single structured risk estimate.

1) Interaction Representation: Every branch below, the safety-time model, the critical-event model, and the bounded residual introduced separately, reads from a single shared representation of the interaction between the ego and agent j. For ego trajectory $\boldsymbol{\xi}_e$ and agent j, the model receives three input groups: current relative kinematics $\boldsymbol{x}^j_{\mathrm{kin}}$ (relative position and velocity, relative heading, closing speed, and each party's own speed at the current instant), the observed interaction sequence $\boldsymbol{X}^j_{\mathrm{int}}$ (relative 3D motion together with a signed-clearance proxy, the center-to-center distance offset by the sum of each box's half-diagonal, sampled over the full observed horizon $[0, H]$), and semantic scene context $\boldsymbol{x}^j_{\mathrm{ctx}}$ (the interaction-type distribution, relative-heading statistics, ego's own heading change, and the agent's object class). The signed-clearance proxy is a lightweight, differentiable stand-in for an exact oriented-box overlap test: cheap to evaluate at every bin and smooth enough to backpropagate through, not the exact geometric quantity a fixed criterion would use.

A bidirectional recurrent encoder (GRU) with attention pooling over $\boldsymbol{X}^j_{\mathrm{int}}$ summarizes the interaction history, while separate MLPs encode $\boldsymbol{x}^j_{\mathrm{kin}}$ and $\boldsymbol{x}^j_{\mathrm{ctx}}$. The three encodings are concatenated, passed through a fusion MLP, and layer-normalized:

$$\boldsymbol{h}^j = f_{\mathrm{enc}}\left(\boldsymbol{x}^j_{\mathrm{kin}}, \boldsymbol{X}^j_{\mathrm{int}}, \boldsymbol{x}^j_{\mathrm{ctx}}\right). \tag{4}$$

Normalizing this final representation keeps its scale fixed regardless of what the rest of the encoder settles into during training. Without it, the single-scalar heads introduced next receive an input whose scale drifts over training and can saturate the bounded nonlinearities they are passed through, silently collapsing their dependence on context.

2) Physically Anchored Safety-Time Modeling: The first branch answers how much recovery margin the interaction with agent j actually requires. It starts from a closed-form physics prior,

$$\tau^j_{\text{phy}} = \text{clip}\left(t_{\text{react}} + \frac{v^j_{\text{close}}}{a_{\text{avail}}} + \epsilon_{\text{unc}},\ [\tau_{\min}, \tau_{\max}] \right), \quad (5)$$

where v^j_{close} is the closing-speed component along the current relative bearing, t_{react} a fixed reaction time, a_{avail} an assumed available deceleration, and ϵ_{unc} a constant pad absorbing prediction uncertainty. All four constants, and the clipping range, are specific to agent j's object class: a pedestrian's evasive capability, reaction time, and predictability are nothing like a vehicle's, so τ^j_{phy} already reflects what kind of agent is being reasoned about before any learning takes place.

CARM corrects this prior in log-space by a bounded, learned amount,

$$\begin{aligned} \mu^j_s &= \log \tau^j_{\text{phy}} + \underbrace{\Delta_{\max} \tanh\big(g_\mu(\boldsymbol{h}^j)\big)}_{\Delta^j_\theta}, \\ \sigma^j_s &= \sigma_{\min} + (\sigma_{\max} - \sigma_{\min}) \sigma\big(g_\sigma(\boldsymbol{h}^j)\big), \\ T^j_s &\sim \text{LogNormal}\big(\mu^j_s, (\sigma^j_s)^2\big), \end{aligned} \quad (6)$$

and treats the required safety time as the random variable T^j_s rather than a single number. Because $\exp(\mu^j_s)$ is exactly the median of a log-normal distribution, the bound $\Delta_{\max} = \log 2$ constrains the median of T^j_s, not its mean, to lie within a factor of two of τ^j_{phy} in either direction: CARM can judge that half the physically assumed margin already suffices, or that twice as much is warranted, but nothing more extreme. τ^j_{phy} is evaluated at every bin edge b_k of the same time bins $0 = b_0 < b_1 < \cdots < b_K = H$ used throughout, rather than as a single scalar, so μ^j_s is itself bin-indexed even though the correction Δ^j_θ is a single scalar per agent-instance applied uniformly across every bin.

3) Critical-Event-Time Modeling: The second branch answers when the interaction with agent j turns critical: the term the overview denotes c^j_θ. Rather than a single scalar, CARM predicts a full discrete-time hazard distribution over the same K bins. For each bin k, a head produces a conditional hazard, $q^j_k = \sigma(g^{\text{evt}}_k(\boldsymbol{h}^j))$, the probability that the critical event falls in bin k given that it has not already occurred in an earlier bin. The event probability mass and cumulative distribution follow the standard discrete-time survival parameterization,

$$p^j_k = q^j_k \prod_{\ell<k} (1 - q^j_\ell), \quad F^j_k = \sum_{\ell=1}^{k} p^j_\ell, \quad (7)$$

so F^j_k, evaluated across all k, is exactly c^j_θ: the probability that the interaction has already turned critical by bin k. Because $\sum_{k=1}^{K} p^j_k = 1 - \prod_{\ell=1}^{K}(1 - q^j_\ell)$, the complementary probability $\prod_{\ell=1}^{K}(1-q^j_\ell) = 1 - F^j_K$ is exactly the probability that no critical event occurs anywhere within the observed horizon, so a genuinely safe interaction is represented on its own terms rather than forced into an artificial event bin. This prediction has no closed-form anchor of its own; how its training label is constructed, independently of the safety-time branch above, is addressed separately.

4) Structured Risk Composition: The two branches combine into $\widetilde{R}^j_{\text{CARM}} = \Gamma(\tau^j_{\text{phy}} + \Delta^j_\theta,\ c^j_\theta)$ from Eq. (3). Writing Φ for the standard normal CDF, the safety-time distribution assigns each bin a probability mass

$$\begin{aligned} \Pr(b_{k-1} < T^j_s \le b_k) &= \Phi\Big(\tfrac{\log b_k - \mu^j_s}{\sigma^j_s}\Big) - \Phi\Big(\tfrac{\log b_{k-1} - \mu^j_s}{\sigma^j_s}\Big), \\ \Pr(T^j_s > b_K) &= 1 - \Phi\Big(\tfrac{\log b_K - \mu^j_s}{\sigma^j_s}\Big), \end{aligned} \quad (8)$$

using μ^j_s, σ^j_s at their own bin from Eq. (6). Because τ^j_{phy}, and therefore μ^j_s, is itself evaluated per bin, each of these $K+1$ quantities is a CDF difference of a distribution parameterized *at that bin*, not of one distribution shared across every bin: each is individually a valid probability in $[0, 1]$, but, unlike disjoint slices of a single CDF, they are not guaranteed to sum to exactly one. Γ combines them with the critical-event probability F^j_k as the value attached to each bin, approximating the expectation of the critical-event CDF at the time the interaction's safety margin runs out,

$$\begin{aligned} R^j_{\text{CARM}} &\approx \mathbb{E}_{T^j_s}\left[F^j_{\text{evt}}(T^j_s) \right] \\ &= \sum_{k=1}^{K} F^j_k \Pr(b_{k-1} < T^j_s \le b_k) + F^j_K \Pr(T^j_s > b_K), \end{aligned} \quad (9)$$

weighted more heavily wherever the local safety-time distribution places more probability mass. Because the weights need not sum to one, R^j_{CARM} is not itself guaranteed to land in $[0, 1]$; a numerical clip is what actually enforces it,

$$\widetilde{R}^j_{\text{CARM}} = \text{clip}\Big(R^j_{\text{CARM}},\ \epsilon,\ 1 - \epsilon \Big), \quad (10)$$

giving the structured risk estimate used throughout, at the cost of an approximation that is exact only where τ^j_{phy} changes little from one bin to the next. Its continuous value is an interaction-level risk, not a direct collision classifier, and keeps the metric able to distinguish near-boundary, collision-free interactions from ones that were never close at all.

C. Bounded Residual Architecture

The residual reads from the same shared interaction representation $\boldsymbol{h}^j$ introduced in Sec. III-B, frozen at this stage. The structured estimate $\widetilde{R}^j_{\text{CARM}}$ combines only two families of distributions, a log-normal safety-time model and a discrete-time hazard model, and this two-distribution proxy need not capture every aspect of the observed margin exactly. CARM accounts for the gap with the learned, bounded correction term already introduced in Eq. (2), named and fixed here as

$$x^j_\theta := \beta \tanh\big(g_r(\boldsymbol{h}^j)\big), \qquad \beta = 2.0, \quad (11)$$

added directly to the risk logit of $\widetilde{R}^j_{\text{CARM}}$ rather than to any physical quantity, to obtain R^j. This is a different correction from Δ^j_θ in an important sense, not just a different

location: Δ_θ^j corrects a time value, τ_{phy}^j, in log-space, before it ever becomes a probability, while x_θ^j corrects the already-composed risk itself, in logit space, after every structured reasoning step has already been applied. It is the only one of CARM's three learned corrections with no closed-form quantity of its own to stay anchored to, so its bound is what alone keeps R^j from drifting arbitrarily far from $\widetilde{R}_{\text{CARM}}^j$. The output $R^j \in [0,1]$ is an interaction-level risk, not a direct collision classifier: its continuous value keeps the metric able to distinguish near-boundary, collision-free interactions from ones that were never close at all.

Score Bounds and Ranking Stability. Because $x_\theta^j \in [-\beta, \beta]$, the residual can flip the ranking between two candidate risks only when their structured estimates $\widetilde{R}_{\text{CARM}}^j$ are already close: whenever the corresponding logit gap exceeds 2β, the final ranking under R^j is guaranteed to match the ranking under $\widetilde{R}_{\text{CARM}}^j$ regardless of what the residual does.

D. Trajectory Risk Aggregation

CARM scores every relevant agent observed alongside the ego trajectory. Each agent's risk R^j is produced independently by the same shared-weight encoder and heads, conditioned only on that agent's own kinematics, interaction history, and context; no other agent's state enters its computation. Agent risks are then combined by the noisy-OR already given in Eq. (2). Because each factor $1 - R^j \in (0,1]$, adding another agent can only shrink the product and therefore raise R: risk accumulates across interactions rather than being dominated by a single worst-case agent, so the number of agents present is itself part of what the score reflects, exactly as a trajectory genuinely becomes riskier the more distinct hazards it faces at once. Because $\boldsymbol{\xi}_e$ has a single, already-realized future, no expectation over multiple predicted future modes is required anywhere in this aggregation.

E. Supervision and Two-Stage Training

Every learned quantity in Secs. III-B and III-C exists to make FutureRisk approximate a target that itself requires no learning at all:

$$R_\theta = f_{\text{FR}}(\boldsymbol{\xi}_e, \mathcal{W}^{\text{obs}}) \approx y. \tag{12}$$

This subsection derives that target y, defines the independent label that supervises the critical-event branch, and describes the two-stage schedule that trains every piece above.

1) Geometric Risk Target: Let $\mathcal{B}_e(t)$ and $\mathcal{B}_j(t)$ denote the oriented 3D occupied volumes of the ego and agent j at time t, read directly from $\boldsymbol{\xi}_e$ and the observed trajectory of agent j in $\mathcal{W}^{\text{obs}}$: the trajectory's own already-realized future, not a forward projection from the current state. We define their geometry-aware 3D time-to-collision as the earliest future overlap within the observed horizon,

$$\tau_{\text{3D}}^j(t) = \inf\{\delta \geq 0 : \mathcal{B}_e(t+\delta) \cap \mathcal{B}_j(t+\delta) \neq \varnothing,\ t+\delta \leq H\}, \tag{13}$$

with $\tau_{\text{3D}}^j(t) = +\infty$ if no overlap occurs within $[t, H]$. Because $\mathcal{B}_e$ and $\mathcal{B}_j$ are read from what each party actually did rather than extrapolated forward from the current state under an assumed, held-constant control input, τ_{3D}^j carries no motion-extrapolation assumption at all, unlike a forward-projected TTC baseline; it can be evaluated directly on the observed future without any prediction step, and reasons over occupied volumes rather than a fixed longitudinal axis, giving a common safety quantity for rear-end, lateral, crossing, and head-on interactions alike.

The target compares τ_{3D}^j against a safety-time requirement $\tau_{\text{safe}}^j(t)$, but here held fixed rather than context-adaptive: the same closed-form physics prior CARM's safety-time branch corrects, $\tau_{\text{safe}}^j(t) = \tau_{\text{phy}}^j(t)$ from Eq. (5), with no learned correction Δ_θ^j applied. This is deliberate: letting the supervision target depend on the very correction it trains would make the target move under the model during training rather than stay fixed. The pairwise and trajectory-level safety margins are then

$$m^j(t) = \tau_{\text{3D}}^j(t) - \tau_{\text{safe}}^j(t), \qquad m = \min_j \min_{t\in[0,H]} m^j(t), \tag{14}$$

positive throughout when the observed trajectory stayed outside this fixed boundary at every instant, negative as soon as one observed interaction crossed it. Mapping the margin through the same monotone transition used throughout,

$$r(t) = \phi\Big(\min_j m^j(t)\Big) = \sigma\Big(-\frac{\min_j m^j(t)}{T_m}\Big) \in [0,1], \tag{15}$$

traces how closely the trajectory approaches this fixed boundary throughout the interaction; its peak value is the trajectory-level analytic target,

$$y = \max_{t\in[0,H]} r(t) = \phi(m). \tag{16}$$

Every quantity feeding y is computed directly from $\boldsymbol{\xi}_e$ and $\mathcal{W}^{\text{obs}}$ using only the fixed prior τ_{phy}^j, so it requires no learned component, no label, and no re-computation as training proceeds; it is what supervises Eq. (12) and what Sec. IV reports as the analytic-only baseline y_{analytic}.

y has a sharp limitation by construction: $\tau_{\text{3D}}^j(t)$ is finite only when an actual box overlap is found, so $m = +\infty$ and $y = 0$ for every trajectory in which no overlap ever occurs, regardless of how close a near-miss came. A routine pass and a narrowly avoided collision are therefore indistinguishable under y alone. This is a deliberate limitation of a purely geometric target, not an oversight: it is exactly what the critical-event branch, supervised next from continuous kinematics rather than binary overlap, exists to correct.

When a dataset additionally carries native, sensor-independent collision ground truth, the value of y used to train CARM on it is further overridden wherever that ground truth disagrees with the geometric margin, since a geometric overlap test can miss a real collision that native sensing confirms:

$$y \leftarrow \begin{cases} 1, & \text{native ground truth confirms a collision,} \\ y, & \text{otherwise,} \end{cases} \tag{17}$$

keeping the continuous, margin-derived value wherever no collision is confirmed, since a binary native label alone

carries no information about how close a non-colliding trajectory came.

2) Critical-Event Supervision: The critical-event branch is supervised independently of the target above, using its own label derived purely from kinematics. Let $\mathrm{sev}(a_{\mathrm{lon}}, a_{\mathrm{lat}}) = \sqrt{\max(-a_{\mathrm{lon}}, 0)^2 + a_{\mathrm{lat}}^2}$ denote a party's evasive-severity score, where a_{lon} is longitudinal (braking) deceleration and a_{lat} is lateral acceleration; we evaluate this once from the ego's own kinematics and once from agent j's own kinematics and take the larger of the two,

$$a^j(t) = \max\big(\mathrm{sev}(a^e_{\mathrm{lon}}(t), a^e_{\mathrm{lat}}(t)),\, \mathrm{sev}(a^j_{\mathrm{lon}}(t), a^j_{\mathrm{lat}}(t))\big),$$

so that a harsh maneuver by either party marks the interaction as safety-critical. A critical event is then the first time this signal exceeds a threshold $\theta_{d,c}$ calibrated for the current dataset d and interaction type $c = c(t, j)$, or an actual collision occurs:

$$T^j_{\mathrm{evt}} = \inf\big\{t \in [0, H] : \mathrm{collision}^j(t) \;\vee\; a^j(t) > \theta_{d,c(t,j)}\big\}, \tag{18}$$

with $T^j_{\mathrm{evt}} = +\infty$ when neither condition is ever met within H. Each $\theta_{d,c}$ is the 95th percentile of $a(t)$ observed for interaction type c within dataset d only, with types below a minimum sample count merged into a catch-all *other* category rather than fit on too few examples: thresholds are fit independently per dataset, never pooled across datasets, so that one dataset's driving style or sensing characteristics cannot distort another's notion of a critical event. Because $\tau^j_{\mathrm{safe}}(t)$ is exactly the quantity the safety-time branch is learning to predict, labeling a critical event by its own margin crossing zero would make its supervision circular; deriving T^j_{evt} from $a(t)$ instead keeps the event target encoding *when* the observed interaction becomes unsafe independently of the safety-time branch's own prediction.

Writing $k_j \in \{1, \ldots, K\}$ for the bin containing T^j_{evt} (bin K standing in for a censored agent with no event inside the horizon), only bins up to and including k_j remain at risk, since the interaction's fate within the horizon is already decided beyond it. Writing $e^j = \mathbb{1}[T^j_{\mathrm{evt}} < \infty]$ for whether an event actually occurred, so $e^j = 0$ for a censored agent regardless of k_j,

$$\mathcal{L}_{\mathrm{evt}} = -\frac{1}{N}\sum_j \sum_{k=1}^{k_j} \Big[w_e \cdot e^j \mathbb{1}[k{=}k_j] \log q^j_k + \big(1 - e^j \mathbb{1}[k{=}k_j]\big) \log(1 - q^j_k) \Big], \tag{19}$$

so a censored agent contributes only negative terms across every at-risk bin up to K, correctly reinforcing low hazard throughout rather than being trained to predict an event in the final bin. with a positive-class weight w_e, the inverse frequency of agents with a finite T^j_{evt} computed once from the training split: a critical event is a small minority outcome, and an unweighted loss would be pulled toward predicting near-zero hazard almost everywhere.

3) Training Objectives and Schedule: The final score is supervised directly against the target from Eq. (17),

$$\mathcal{L}_{\mathrm{risk}} = \mathrm{BCE}(R, y), \tag{20}$$

again with its own inverse-frequency positive-class weight, over instances with $y > 0.5$, computed once from the training split for the same reason as w_e above. The safety-time branch receives no direct regression supervision of its own: since $\tau^j_{\mathrm{safe}}(t)$ has no independently observable ground truth, constructing a pseudo-label for T^j_s and regressing against it would make the supervision circular. Instead, T^j_s is shaped entirely through its role inside the composition in Eq. (9), so gradients into μ^j_s, σ^j_s come solely from $\mathcal{L}_{\mathrm{risk}}$. We summarize the training objective as

$$\mathcal{L} = \lambda_e \mathcal{L}_{\mathrm{evt}} + \lambda_r \mathcal{L}_{\mathrm{risk}}. \tag{21}$$

Training uses two stages. The first stage disables the residual ($x^j_\theta \equiv 0$) and jointly learns the encoder, the critical-event branch, and the safety-time branch using $\lambda_e \mathcal{L}_{\mathrm{evt}} + \lambda_r \mathcal{L}_{\mathrm{risk}}$, together with a small penalty discouraging g_μ, g_σ's raw outputs from growing large enough to saturate the bounding nonlinearities they are passed through: neither head receives any direct supervision of its own, so nothing else in the objective keeps them well scaled. The second stage freezes this structured branch entirely and trains only the bounded residual, using $\lambda_r \mathcal{L}_{\mathrm{risk}}$ alone. This separation lets CARM improve scalar calibration without letting the free residual distort the two interpretable, boundary-anchored latent distributions underneath it.

IV. Experiments

Setup. An instance is one ego trajectory with its surrounding agents over an $H = 4\,\mathrm{s}$ window ($K = 20$ bins). In-domain results use the held-out test split of DeepAccident [13] (train/val/test 4967/1656/1656), which provides native, sensor-independent collision labels. Cross-domain results apply the same DeepAccident-trained CARM to all 3679 Bench2Drive [15] instances without any parameter fit on that dataset. Bench2Drive has no native collision annotation, so its collision label is our geometric 3D-TTC overlap test (Eq. (13)); this makes the cross-domain label circular for any method that is itself a function of that test, which we mark in the tables, and we base claims about collision agreement on the in-domain, native-label results. CARM uses width 64 and dropout 0.1 throughout, trained with Adam (learning rate 10^{-3}, weight decay 10^{-4}, batch size 32) for 40 Stage-1 and 20 Stage-2 epochs; all numbers are from a single training seed unless stated otherwise.

Baselines and metrics. Rule-based baselines are the analytic target y_{analytic} (Eq. (16)), raw 3D-TTC, PET, DRAC, the RSS safety envelope [11], and Forward TTC, our instantiation of Li *et al.*'s analytical framework [12], which projects each party forward from its current state under a held-constant control input rather than reading the observed future; we report its 2-D (constant turn rate and velocity) and 1-D (no turning) forms. Learning-based baselines are Lu *et al.*'s unsupervised surrogate-safety embedding [10], retrained on the DeepAccident training split, and GSSM [16] with its released naturalistic-driving checkpoint. Agreement with collision outcomes is measured by AUROC and AUPRC

TABLE I

INSTANCE-LEVEL DISCRIMINATION AGAINST THE BINARY COLLISION LABEL, DEEPACCIDENT.

Method	AUROC ↑	AUPRC ↑
Rule-based		
y_{analytic}	0.648	0.372
raw 3D-TTC	0.646	0.350
Forward TTC (2-D) [12]	0.685	0.425
Forward TTC (1-D)	0.684	0.424
RSS [11]	0.529	0.199
Learning-based		
Lu *et al.* [10]	0.662	0.452
GSSM [16]	0.567	0.226
Our method		
FutureRisk (CARM)	**0.793**	**0.490**

TABLE II

FINE-GRAINED DISCRIMINATION, DEEPACCIDENT. ρ VS. m: $n = 286$, FAR FEWER THAN $a(t)$'S $n = 6630$.

Method	ρ vs. $a(t)$	ρ vs. m
Rule-based		
y_{analytic}	0.007	1.000 *(circ.)*
raw 3D-TTC	0.006	– (const.)
Forward TTC (2-D) [12]	0.274	– (const.)
Forward TTC (1-D)	0.265	– (const.)
PET	−0.093	0.155
DRAC	0.245	0.558
RSS [11]	**0.363**	0.122
Learning-based		
Lu *et al.* [10]	−0.001	0.430
GSSM [16]	0.070	0.080 *(n.s.)*
Our method		
FutureRisk (CARM)	**0.276**	0.290

against the binary collision label. Fine-grained discrimination is measured on collision-free instances by Spearman's ρ against two references: the evasive-severity proxy $a(t)$ (Eq. (18)), available for every such instance, and the worst-case geometric margin m (Eq. (14)), finite only when an overlap exists within the horizon and thus available for far fewer. Planner comparisons use matched scenarios with the surrounding agents fixed (Sec. III-A).

A. Consistency with Collision-Based Evaluation

Table I reports AUROC and AUPRC against DeepAccident's native collision label. FutureRisk reaches 0.793/0.490, ahead of every baseline on both metrics; the strongest baselines are Forward TTC (2-D) on AUROC (0.685) and Lu *et al.* on AUPRC (0.452).

The rule-based baselines separate by what they read. y_{analytic} and raw 3D-TTC read the observed future through a discrete-frame box overlap test, which can miss contacts that the native label records (Sec. III-E); both stay near 0.65. Forward TTC lands slightly higher, plausibly because a continuous projection is more forgiving of near-contact than an exact overlap test; its 2-D and 1-D forms are indistinguishable here. RSS is lowest (0.529/0.199): a fixed envelope over current speed and gap is too conservative and scene-agnostic to track which interactions end in contact. Among learning-based baselines, Lu *et al.*'s sequence embedding gives the best baseline AUPRC but a middling AUROC (0.662), and GSSM, applied zero-shot from a naturalistic-driving checkpoint, falls below every TTC-based baseline and above only RSS (0.567/0.226). FutureRisk's margin over all of them comes from structure no baseline has: a safety-time distribution composed with an independently supervised critical-event distribution. Sec. IV-C attributes the gain to each learned term.

B. Fine-Grained Discrimination

Table II asks whether each score can order collision-free trajectories by how close they came. FutureRisk correlates positively with both references: $\rho = 0.276$ against the evasive-severity proxy $a(t)$ ($n = 6630$) and $\rho = 0.290$ against the worst-case geometric margin m ($n = 286$). Fig. 6 shows the ordering is graded rather than thresholded: both the mean and the median of R increase across the five $a(t)$ quintiles, including the lower ones that lie well below the 95th-percentile threshold used to construct the critical-event label (Eq. (18)).

The two references favor different hand-crafted quantities, and the pattern in Table II follows from what each is built from. $a(t)$ is a function of the parties' current accelerations, so measures computed from current speed and gap track it closely by construction (RSS 0.363, Forward TTC 0.274), while measures that read the already-evaded future are near zero (y_{analytic} 0.007, raw 3D-TTC 0.006). m exists only where an overlap occurred; on that subset y_{analytic} is $\phi(m)$ itself (marked *circ.*), raw 3D-TTC and Forward TTC collapse to a single value, and DRAC, which directly encodes the deceleration required to avoid contact, tracks it most tightly (0.558). Each hand-crafted quantity is strong on the reference it is derived from and weak on the other: RSS falls to 0.122 against m, DRAC to 0.245 against $a(t)$. The learned embedding of Lu *et al.* shows the same split (0.430 against m, −0.001 against $a(t)$). FutureRisk is the only score in Table II whose correlation exceeds 0.25 against both references, and it is also the score that leads on collision agreement (Sec. IV-A); no other method offers this combination.

C. Interpretability–Adaptivity Trade-off

We quantify how anchored a score is as the fraction of its final logit contributed by the bounded structured branch rather than the free residual: 100% for RSS and 0% for Lu *et al.* by construction – these are exact structural facts, not estimates, since neither method has any free component (RSS) or any closed-form component (Lu *et al.*) to begin with – and $1 - \mathbb{E}\big[\,|x_\theta|/|\operatorname{logit} \widetilde{R}_{\text{CARM}}|\,\big]$ for FutureRisk, which *is* an empirical estimate, computed per agent and averaged over the DeepAccident test set. GSSM cannot be placed on this axis at all: it predicts $\log(\text{spacing}) \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2)$ directly from an MLP over raw kinematic features, with no closed-form quantity ever mixed additively into its output the way $\log \tau_{\text{phy}}$ is mixed into μ_s (Eq. (6)) – there is no structured/free split in its computation graph to measure a fraction of, so we mark it *n/a* in Table IV rather than force a number. This is itself evidence for the trade-off FutureRisk claims to resolve: of the four methods compared, FutureRisk

is the only one whose position on the interpretability–adaptivity axis is even *measurable* on the same scale as its discriminative power, because it is the only one built with a closed-form branch to measure a fraction of in the first place.

Table IV places all four side by side. FutureRisk keeps 70.1% of its logit anchored (mean) while leading on collision agreement both in-domain (0.793) and cross-domain (0.827; Lu *et al.*'s 0.990 there is circular, since its input TTC sequence is the quantity the cross-domain label is built from). RSS, fully anchored, does not track collisions (0.529); Lu *et al.*, fully free, does not track evasive severity (-0.001); GSSM, also fully free but conditioned on richer kinematic context than Lu *et al.*'s TTC sequence, does better on both AUROC columns (0.567/0.817) but still trails FutureRisk, and its correlation with evasive severity is weak (0.070). FutureRisk fails on neither axis.

The 70.1% figure is a mean over $75{,}752$ scored agents (DeepAccident in-domain test set), and Fig. 2 shows it is not an average pulled between a bimodal population of "fully anchored" and "fully free" agents: the distribution is unimodal, with median 0.741 (IQR 0.666–0.803) in-domain and median 0.846 (IQR 0.805–0.878) cross-domain (Bench2Drive) – the median sits *above* the reported mean because a small tail (0.21% of in-domain agents, 0.19% cross-domain) has $\text{logit}\,\widetilde{R}_{\text{CARM}} \approx 0$ (the structured branch alone is at its own decision boundary), where this ratio-based estimator is not numerically well-behaved; excluding that tail moves the mean from 0.701 to 0.720 in-domain without moving the median. Grouping by whether an agent's own ground-truth critical-event label T_{evt} (Eq. (18)) is finite within the horizon – not a threshold on the model's own belief – separates the distribution more than the binary collision label does: median anchored fraction is 0.700 when a critical event actually occurs versus 0.769 when it does not (rank-biserial $r = 0.371$, in-domain), while collision vs. no-collision shifts the median by only 0.009 ($r = -0.066$, statistically significant at $n = 75{,}752$ but not a practically meaningful separation). The model uses more of its free residual specifically on interactions that turn out to be genuinely critical, not on interactions that merely end in contact – consistent with a model that mostly does not need to correct its physical prior, and reaches for the free residual only where the prior's assumptions are under the most strain. All of this comes from one training seed; the box plot's uncertainty is a bootstrap over test agents *within* that seed (sampling uncertainty of the estimate), not a training-seed stability result, which this project has not yet run for any CARM configuration.

Table III decomposes R by adding the learned terms of Eqs. (6)–(11) back one at a time. The structure alone, with τ_{phy} uncorrected and no residual, already reaches 0.731 in-domain AUROC, above every baseline in Table I, and 0.799 cross-domain: the composition of a safety-time distribution with a critical-event distribution clears prior work before any learned correction is applied. The median shift Δ_θ adds 0.022 in-domain AUROC on its own; the spread σ_s helps only in combination with it (Stage-1, 0.764), not in isolation (0.718), so the two are used together. The Stage-2 residual x_θ contributes the largest single increment ($0.764 \to 0.793$) and gives the best value in all four columns. Along the training path actually used, prior only $\to$ Stage-1 $\to$ full CARM, three of the four columns improve monotonically; cross-domain AUPRC dips at Stage-1 ($0.344 \to 0.333$) and recovers with the residual (0.354). Because $\Delta_{\max}$ bounds the median shift to a factor of two and β bounds the residual to $[-2, 2]$ in logit space, these gains are obtained without letting the score leave the neighborhood of the closed-form prior.

TABLE III

FULL DECOMPOSITION ABLATION OF R: EVERY LEARNED TERM ADDED BACK ONE AT A TIME (SINGLE SEED).

Model	AUROC(in)	AUPRC(in)	AUROC(cross)	AUPRC(cross)
raw (τ_{phy} only)	0.731	0.442	0.799	0.344
$+\,\Delta_\theta$ only	0.753	0.483	0.812	0.341
$+\,\sigma_s$ only	0.718	0.417	0.802	0.331
Stage-1 only (both, $g_r \equiv 0$)	0.764	0.459	0.818	0.333
Full CARM	**0.793**	**0.490**	**0.827**	**0.354**

TABLE IV

INTERPRETABILITY–ADAPTIVITY TRADE-OFF (SINGLE SEED). ANCHORED: 100%/0% ARE EXACT STRUCTURAL FACTS (RSS HAS NO FREE COMPONENT, LU *et al.* HAS NO CLOSED-FORM COMPONENT), 70.1% IS AN EMPIRICAL MEAN OVER PER-AGENT ESTIMATES (FIG. 2 GIVES ITS DISTRIBUTION); GSSM'S IS MARKED N/A BECAUSE IT HAS NO CLOSED-FORM/FREE DECOMPOSITION TO MEASURE A FRACTION OF (SEE TEXT).

Method	Anchored ↑	AUROC(in)	AUROC(cross)	ρ vs. $a(t)$
RSS [11]	100%	0.529	0.662	0.363
Lu *et al.* [10]	0%	0.662	0.990(*circ.*)	−0.001
GSSM [16]	n/a	0.567	0.817	0.070
FutureRisk (CARM)	70.1%	**0.793**	**0.827**	**0.276**

D. Planner Comparison and Qualitative Analysis

We test whether FutureRisk can rank planners the way collision rate does, but with finer resolution. Four released Bench2Drive planners [14] are used: AD-MLP (kinematics only), TCP (single front camera), and UniAD-Tiny and VAD-Base (surround-view BEV). On each of $n = 15$ eligible *AccidentTwoWays* clips, the surrounding agents' tracks are held fixed and only the ego box is replaced by the planner's predicted path, interpolated onto CARM's time grid and extrapolated at constant velocity beyond the planner's own horizon. All trajectories are scored zero-shot by the DeepAccident-trained checkpoint. Collision rate here means the fraction of clips with a geometric overlap (Eq. (13)).

Fig. 3 compares mean R with collision rate per planner. The two agree on TCP (safest) and AD-MLP (second), and disagree on the remaining pair: mean R ranks UniAD safer than VAD, collision rate ranks VAD safer than UniAD. With $n = 15$ the aggregate gap is small (0.382 vs. 0.423), and Fig. 4 locates almost all of it in a single clip where the two metrics disagree by construction. In that clip, VAD's trajectory never overlaps any agent's box within the 4 s

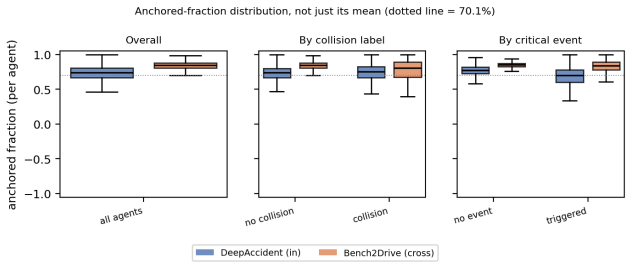


Fig. 2. Per-agent anchored-fraction distribution (DeepAccident in-domain, Bench2Drive cross-domain; single seed), split by native/geometric collision label and by whether the agent's ground-truth critical-event label T_{evt} is finite within the horizon. Dotted line: the 70.1% mean from Table IV. Boxes clipped to $[-1, 1]$; outliers below -1 ($< 0.22\%$ of agents, a ratio-blowup artifact discussed in the text) are excluded from the plot but not from any reported statistic.

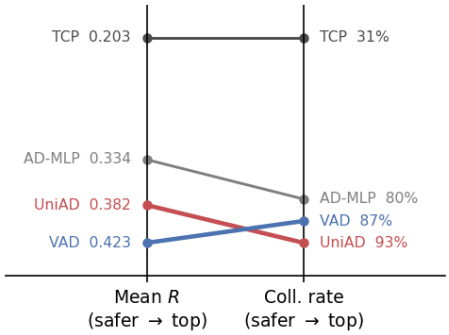


Fig. 3. Planner ranking by mean R (left) and by geometric collision rate (right) over the same $n = 15$ clips; on both axes, safer is higher. Parallel lines (TCP, AD-MLP) mean the two metrics agree on the order; crossing lines (UniAD, VAD) mean they disagree.

window, so the overlap test and the closed-form margin register nothing ($m(t) = +\infty$, $r(t) \equiv 0$, panel (a)). Its distance to one agent nevertheless shrinks monotonically from 57.8 m to 3.4 m and is still closing at about 4 m/s when the window ends (panel (b2)); the critical-event hazard q_k for that agent rises steeply over the final bins (panel (c)) and the trajectory scores $R = 0.975$. UniAD, in the same scenario, collides with a different agent at $t = 0.6$ s (panel (b1)) and scores $R = 0.962$. Removing this clip brings VAD's mean R to 0.383, level with UniAD's 0.382: the planner-level disagreement is not noise but one interaction that a fixed-horizon overlap test cannot register until contact occurs. This is the case the critical-event branch was built for (Sec. III-E): the geometric target y is 0 by construction whenever no overlap occurs, and it is the independently supervised hazard q_k, read from the realized interaction rather than from any overlap, that carries the signal here.

Fig. 5 shows per-frame scores along the ego path for two DeepAccident instances: a near-miss (top; no contact) and a collision that the geometric overlap test misses entirely (bottom; native label confirms contact at $t = 4.0$ s). On the missed collision, raw 3D-TTC is 0.0 at all 41 frames, the same fixed-horizon blindness seen for VAD above. RSS and GSSM respond, but at other times (RSS peaks at 0.552 at $t = 3.1$ s, GSSM at 3.85 at $t = 3.6$ s, in its own unbounded units). FutureRisk is elevated at the confirmed contact frame ($R = 0.478$) and crosses 0.4 XX frames before contact. Its scale is bounded to $[0, 1]$ and, through the anchoring of Sec. IV-C, stays tied to the physics prior, which neither GSSM's unbounded intensity nor RSS's collision-insensitive envelope offers. On the near-miss, all four methods rise toward the closest-approach point without flagging a collision; FutureRisk shows a transient spike at $t = 0.3$ s, early in the clip, and settles to the same near-certain value within one frame. Under the geometric overlap test both instances carry label 0, one of them wrongly; FutureRisk assigns them distinct, correctly ordered scores.

E. Perturbation Sweep and Hyperparameter Sensitivity

Perturbation sweep. Starting from a real DeepAccident collision, we shift the ego path laterally (perpendicular to its per-frame heading) from $\delta = 0$, where it collides, to $\delta = 6$ m, where it passes with ample room, sampling 55 offsets and recomputing R, raw 3D-TTC, and PET at each (Fig. 7). The two rule-based scores are step functions of δ: raw 3D-TTC drops from 1.0 to 0.0 between two adjacent offsets 0.02 m apart at the overlap boundary ($\delta \approx 2.15$ m), and PET does the same later ($\delta \approx 5.1$–5.4 m). R moves smoothly across the whole range; its largest change between adjacent offsets is 0.019 and its total variation is 0.19. What the sweep compares is how far each score moves when the trajectory moves a little: R moves a little, and each rule-based score moves its full scale at one offset.

Hyperparameter sensitivity. The transition width T_m in Eq. (15) rescales the margin monotonically, so it cannot change any ranking of y_{analytic}; we checked this on all 8279 DeepAccident instances (Kendall's $\tau = 1$ between $T_m \in \{0.5, 2.0\}$ and $T_m = 1.0$). The only way T_m reaches CARM is through the training target. When we retrain over $T_m \in \{0.5, 1.0, 2.0\}$ and critical-event percentile $\{90, 95, 99\}$, at a reduced budget of 12 Stage-1 and 6 Stage-2 epochs, in-domain AUROC across the nine settings varies by at most 0.019 (0.735–0.754).

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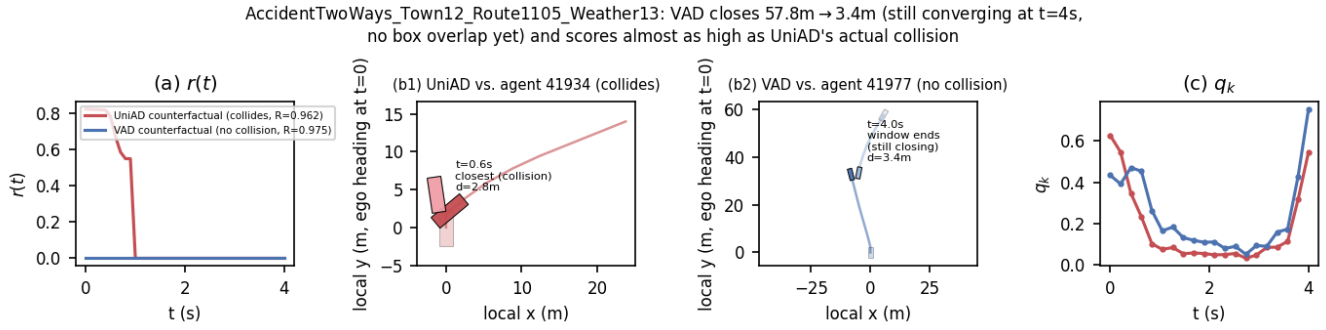


Fig. 4. One AccidentTwoWays scenario (Route 1105) under UniAD (red) and VAD (blue). UniAD collides with one agent at $t = 0.6\,\mathrm{s}$ ($R = 0.962$). VAD never overlaps any agent's box within the 4 s window, yet closes on another agent from 57.8 m to 3.4 m and is still approaching at about 4 m/s when the window ends ($R = 0.975$). (a) Closed-form per-frame risk $r(t)$ (Eq. (15)): nonzero for UniAD, zero for VAD. (b1)/(b2) Bird's-eye view of ego and agent paths in each ego's local frame at $t = 0$; light boxes mark footprints at $t = 0$, solid boxes at the moment of interest (UniAD: contact; VAD: end of window). (c) Critical-event hazard q_k (Eq. (7)) for the agent each case concerns.

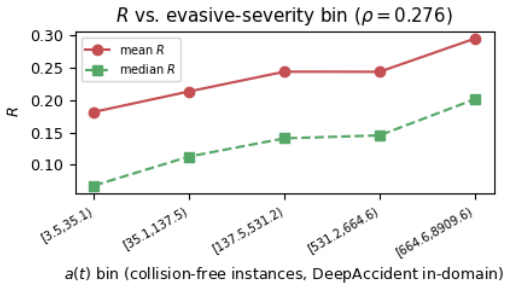


Fig. 6. Mean and median R within each quintile of the evasive-severity proxy $a(t)$, collision-free DeepAccident test instances ($n = 6630$). Both increase with severity; Spearman $\rho = 0.276$ over all instances.

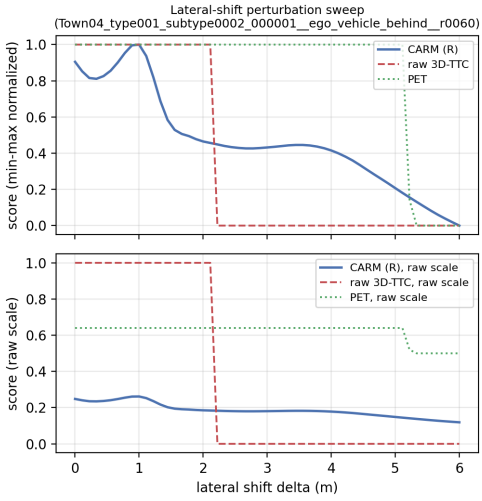


Fig. 7. R, raw 3D-TTC, and PET as the ego path of one DeepAccident collision instance is shifted laterally by δ (top: each score min–max normalized; bottom: raw scale). Raw 3D-TTC and PET each jump from one extreme to the other at a single offset; R changes smoothly throughout.

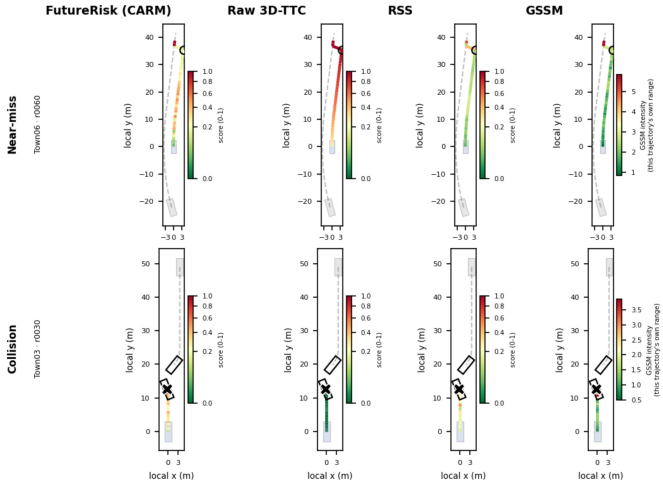


Fig. 5. Per-timestep risk along the ego path for two real DeepAccident instances (rows) under four methods (columns): a near-miss safely avoided (top; native collision label = 0) and a collision missed entirely by the fixed-horizon geometric criterion (bottom; native collision label = 1, contact at $t = 4.0$s). Path color is that method's own per-frame score (green=low, red=high); FutureRisk, RSS, and raw 3D-TTC share one fixed $[0, 1]$ color scale, while GSSM's unbounded intensity is normalized to each row's own observed range (its own colorbar). Gray dashed path is the interacting agent; X marks the native-confirmed collision contact; o marks the closest-approach frame on the near-miss row, where no collision occurs.

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